

Lecture 21

Quantum Calogero–Sutherland: integrability without a Bethe ansatz

Objectives. Every model since [Lecture 15](#) has been solved by some form of Bethe ansatz. This lecture quantises [Lecture 11](#)’s Calogero–Moser gas and none of that machinery appears. The ground state is a product of powers, and one page of differentiation confirms it. The commuting charges come from operators that differentiate and permute at once — Dunkl operators — whose commutativity is a statement about transpositions, not about the Yang–Baxter equation. In a harmonic trap the whole spectrum is an arithmetic progression, the quantum face of [Lecture 11](#)’s isochrony. And what survives of the Bethe ansatz survives as a result, not a method.

21.1 The models, the units, and the domain

[Lecture 11](#) studied N unit masses on a line with the pair potentials $1/x^2$, $1/\sinh^2 x$, $1/\sin^2 x$ and $\wp(x)$ ([Figure 21.1](#)). Two of them are quantised here: the rational one in a harmonic trap, which is [Lecture 11](#)’s isochronous model (§ 11.6), and the trigonometric one, which is Sutherland’s.

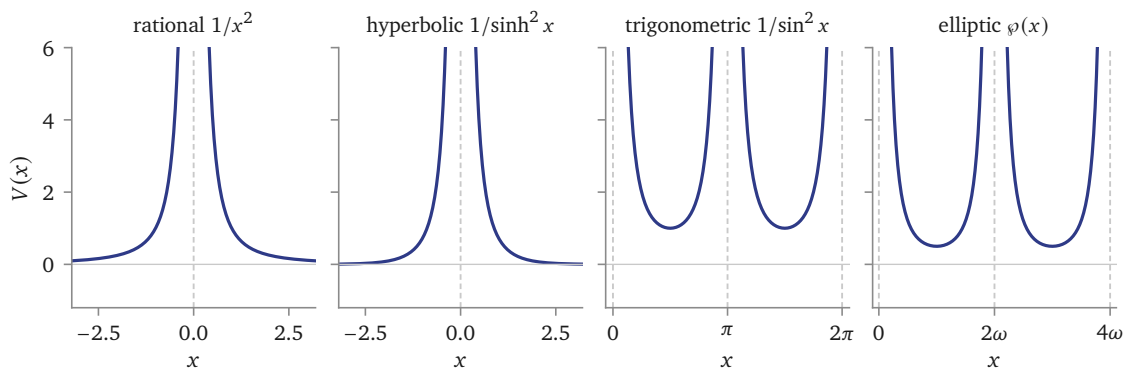


Figure 21.1: [Figure 11.1](#) again, for reference. This lecture quantises the first panel — in a harmonic trap, so that the gas is confined — and the third, on a circle of circumference π . The hyperbolic and elliptic models are quantum-integrable too and are not done here.

Remark 21.1 (Units, and the fourth named clash). This lecture uses $\hbar = m = 1$, kinetic operator $-\frac{1}{2} \sum_j \partial_j^2$, so that [Lecture 11](#)’s $H = \frac{1}{2} \sum p^2 + g^2 \sum V$ is its literal classical limit — and not the $\hbar = 2m = 1$ of [Lecture 16](#), [Lecture 19](#) and [Lecture 20](#), which belongs to Lieb–Liniger.

Mixing the two in one formula is how a model with no Jastrow ground state gets printed. The trap frequency is ν , as in Equation (11.10); ω is the symplectic form and nothing else.

In these units the two Hamiltonians are

$$H_C = -\frac{1}{2} \sum_{j=1}^N \partial_j^2 + \sum_{i<j} \frac{\beta(\beta-1)}{(x_i - x_j)^2} + \frac{\nu^2}{2} \sum_{j=1}^N x_j^2 \quad (21.1)$$

on $\mathbb{R}^N$, and

$$H_S = -\frac{1}{2} \sum_{j=1}^N \partial_j^2 + \sum_{i<j} \frac{\beta(\beta-1)}{\sin^2(x_i - x_j)} \quad (21.2)$$

on a circle of circumference π . With $\hbar$ restored the coupling is $\hbar^2\beta(\beta-1)$, so $\beta = \frac{1}{2} + \sqrt{\frac{1}{4} + g^2/\hbar^2}$ and $\hbar\beta \rightarrow g$ as $\hbar \rightarrow 0$: Lecture 11's g is the classical limit of $\hbar\beta$, and the classical regime is $\beta \rightarrow \infty$ · checked.

Why the coupling is written that way, and what the domain is

Near a collision the relative coordinate $r = x_i - x_j$ sees $-\partial_r^2 + \beta(\beta-1)/r^2$, whose solutions behave as r^p with $p(p-1) = \beta(\beta-1)$: two exponents,

$$p = \beta \quad \text{and} \quad p = 1 - \beta, \quad (21.3)$$

which the coupling does not distinguish · checked. Fix $\beta \geq \frac{1}{2}$, choosing the larger — a labelling convention, not a stability condition, since $\beta(\beta-1) = (\beta - \frac{1}{2})^2 - \frac{1}{4}$ never falls below the Hardy bound $-\frac{1}{4}$, which it attains at $\beta = \frac{1}{2}$, where the exponents coincide and the second solution is $r^{1/2} \ln r$.

Definition 21.2 (The model, completely). Take the ordered chamber — $x_1 < \dots < x_N$ on the line, and $0 \leq x_1 < \dots < x_N < \pi$ on the circle, where $x_N - x_1 \rightarrow \pi$ counts as a collision too — and define the operator as the *Friedrichs extension* of (21.1) or (21.2) from the smooth functions compactly supported away from every collision. On this domain an eigenfunction behaves at a collision as

$$\psi = O(|x_i - x_j|^\beta), \quad x_i \rightarrow x_j, \quad (21.4)$$

the *larger* of the two exponents in (21.3). For $\beta \geq \frac{3}{2}$ the Friedrichs extension is the only self-adjoint one, the other branch $r^{1-\beta}$ being not even locally square-integrable; for $\frac{1}{2} \leq \beta < \frac{3}{2}$ the operator is not essentially self-adjoint, and (21.4) is what picks *this* extension out of the family · cited: *Reed and Simon II, §X.1 and its appendix*. For (21.2) there is a second, global condition: the model is *bosonic and periodic*, so a state extends from the chamber to a symmetric function of period π in each coordinate.

Two conventions follow and are used throughout. Restricting to the chamber and extending bosonically to $\mathbb{R}^N$ are unitarily equivalent up to $\sqrt{N!}$, so a symmetric function on $\mathbb{R}^N$ and a function on the chamber are the same object; and $\Delta = \prod_{i<j} (x_j - x_i)$ is positive there, so the symmetric function Δ^β names is $|\Delta|^\beta$. And (21.4) describes *eigenfunctions*, not the whole operator domain, which contains functions with no pure power behaviour at all.

Example 21.3 ($\beta = 1$: hard-core bosons, and free fermions). At $\beta = 1$ the coupling vanishes but the boundary condition does not: it becomes $\psi = 0$ at every collision. The particles are

free and impenetrable — the Tonks–Girardeau gas of [Proposition 19.7](#), here on a circle rather than on [Lecture 14](#)’s lattice. Girardeau’s map $\psi_B = \prod_{i<j} \text{sgn}(x_i - x_j) \psi_F$ turns a fermionic state into a bosonic one at the same energy, and carrying one particle once round the circle multiplies the sign factor by $(-1)^{N-1}$ · *checked*. So the impenetrable Bose gas on a ring is free fermions with *antiperiodic* boundary conditions when N is even and periodic ones when N is odd, as [Lecture 19](#) said for the continuum gas · *cited: Girardeau (1960)*. For $N = 2$ the allowed momenta on a circle of circumference π are then the odd integers, the two lowest are ± 1 , and $E_0 = 1$. [Theorem 21.13](#) will return that number from a formula that never mentions fermions.

21.2 Jastrow ground states

Write $\Delta = \prod_{i<j} (x_j - x_i)$, positive on the chamber, and

$$\psi_0^C = \Delta^\beta e^{-\nu \sum_j x_j^2/2}, \quad \psi_0^S = \prod_{i<j} (\sin(x_j - x_i))^\beta. \quad (21.5)$$

Both obey (21.4) at every collision — for ψ_0^S including the wrap-around one, since $\sin$ has period π — and both are positive on the chamber.

Theorem 21.4 (The Jastrow states). $H_C \psi_0^C = E_0^C \psi_0^C$ and $H_S \psi_0^S = E_0^S \psi_0^S$ with

$$E_0^C = \nu \left(\frac{N}{2} + \frac{\beta N(N-1)}{2} \right), \quad E_0^S = \frac{\beta^2 N(N^2-1)}{6}. \quad (21.6)$$

Without the trap, $H_C|_{\nu=0} \Delta^\beta = 0$.

Proof. For $\psi = e^W$, $\psi^{-1}(-\frac{1}{2} \sum \partial_j^2) \psi = -\frac{1}{2} \sum_j (\partial_j^2 W + (\partial_j W)^2)$, so everything is an exercise in two identities. Take $W = \beta \sum_{i<j} \ln(x_j - x_i)$ and write $x_{ij} = x_i - x_j$. Then $\partial_i W = \beta \sum_{j \neq i} x_{ij}^{-1}$ and $\partial_i^2 W = -\beta \sum_{j \neq i} x_{ij}^{-2}$, so

$$-\frac{1}{2} \sum_i (\partial_i^2 W + (\partial_i W)^2) = \frac{\beta}{2} \sum_i \sum_{j \neq i} \frac{1}{x_{ij}^2} - \frac{\beta^2}{2} \sum_i \sum_{j \neq i} \frac{1}{x_{ij}^2} - \frac{\beta^2}{2} \sum_i \sum_{\substack{j,k \neq i \\ j \neq k}} \frac{1}{x_{ij} x_{ik}}.$$

The last sum vanishes: each unordered triple $\{i, j, k\}$ contributes six terms, two copies of $(x_{ij} x_{ik})^{-1} + (x_{ji} x_{jk})^{-1} + (x_{ki} x_{kj})^{-1}$, which is zero after clearing denominators. The first two give $-\beta(\beta-1) \sum_{i<j} x_{ij}^{-2}$, which cancels the potential exactly. That is the last claim.

Now add the Gaussian, $W \mapsto W - \nu \sum_j x_j^2/2$. The new terms are $-\frac{1}{2} \sum_i (-\nu) = \nu N/2$, then $-\frac{1}{2} \nu^2 \sum x_j^2$, which cancels the trap, and the cross term

$$\nu \beta \sum_i \sum_{j \neq i} \frac{x_i}{x_{ij}} = \nu \beta \sum_{i<j} \frac{x_i - x_j}{x_{ij}} = \nu \beta \frac{N(N-1)}{2}.$$

Together these are E_0^C .

For ψ_0^S replace x_{ij}^{-1} by $\cot x_{ij}$. Now $\partial_i^2 W = -\beta \sum_{j \neq i} \csc^2 x_{ij}$ and $\cot^2 = \csc^2 - 1$, so the two-index terms again cancel the potential and leave $\beta^2 N(N-1)/2$; and the three-index sum is no longer zero but is fixed by the cotangent identity $\cot a \cot b + \cot b \cot c + \cot c \cot a = 1$ whenever $a + b + c = 0$, applied to $a = x_{ij}$, $b = x_{jk}$, $c = x_{ki}$. Each unordered triple contributes -2 , so that sum is $-N(N-1)(N-2)/3$, and $-\frac{1}{2} \beta^2$ times it is $+\beta^2 N(N-1)(N-2)/6$. Hence

$E_0^S = \beta^2 N(N-1)/2 + \beta^2 N(N-1)(N-2)/6$, which is (21.6) · checked: symbolically at $N = 3$, in both the trigonometric and the z variables, and by exact evaluation at $N = 4, 5, 6$. ■

A trapped Jastrow density is drawn in Figure 21.2. At $\beta = 1$ the function $\Delta e^{-\nu \sum_j x_j^2/2}$ is the Slater determinant of the N lowest oscillator states, so ψ_0^C is exactly the free-fermion ground state — and (21.6) returns $\nu(N/2 + N(N-1)/2) = \nu \sum_{n < N} (n + \frac{1}{2})$, as it must.

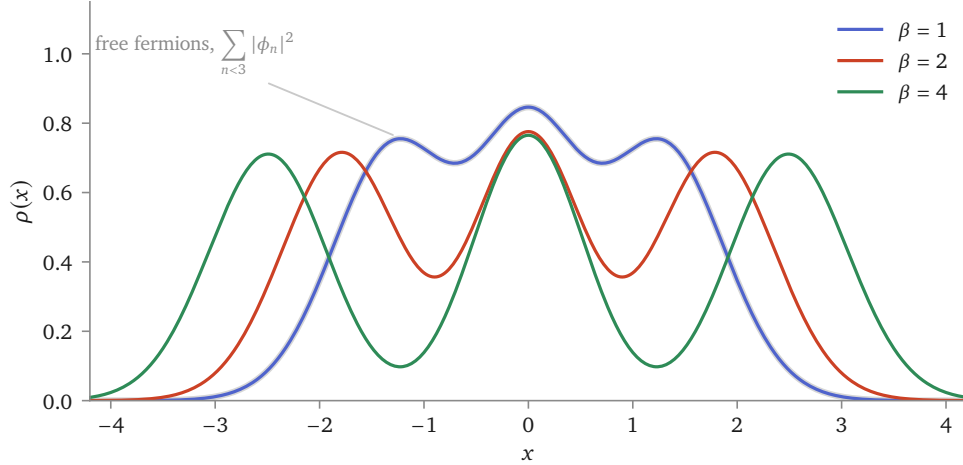


Figure 21.2: The single-particle density of the trapped Jastrow state (21.5) at $N = 3$, for $\beta = 1, 2, 4$ (quadrature, $\nu = 1$). The $\beta = 1$ curve lies on the free-fermion density $\sum_{n < 3} |\phi_n|^2$, in grey behind it, because $\Delta e^{-\nu r^2/2}$ is their Slater determinant. Raising β widens the cloud; the generator asserts all three statements.

Why it is the ground state, and why positivity is not the reason

The usual argument is that a positive eigenfunction of a Schrödinger operator on a connected domain is the ground state. That is true, and here it needs a hypothesis that is easy to drop.

Remark 21.5 (The partner exponent). Replacing β by $1 - \beta$ in ψ_0^S changes neither (21.2) nor the positivity of the function on the chamber, and Theorem 21.4’s computation — which used only the exponent — makes it solve the same differential equation with the value $(1 - \beta)^2 N(N^2 - 1)/6$, strictly smaller than E_0^S for $\beta > \frac{1}{2}$ · checked. So positivity alone cannot identify the ground state. What excludes the partner is Definition 21.2, and the test is sharper than it looks: at $N = 2$ the partner leaves L^2 exactly at $\beta = \frac{3}{2}$, but for $N \geq 3$ the binding collision is the total one, where $\Delta^{1-\beta}$ is homogeneous of degree $(1 - \beta)N(N-1)/2$ in $N-1$ relative coordinates and its square is integrable only for $\beta < 1 + 1/N$. A cluster of m particles gives $\beta < 1 + 1/m$, and $1 + 1/m$ decreases in m , so the total collision is the sharpest of the $N-1$ tests. At $N = 3$, $\beta = \frac{7}{5}$ every pair test passes and the triple collision diverges · checked. So the honest statement is that (21.5) is positive, lies in the form domain of Definition 21.2, and is the ground state of that operator · sketched.

21.3 Dunkl operators

Lecture 11 got the classical charges from a Lax matrix; Lecture 17 got them from a transfer matrix. Neither route is taken here. Instead there is a family of first-order operators that commute outright — and they do it by carrying a transposition inside them.

Definition 21.6 (Dunkl operators). Let s_{ij} be the operator that exchanges the arguments x_i and x_j of a function, $(s_{ij}f)(\dots, x_i, \dots, x_j, \dots) = f(\dots, x_j, \dots, x_i, \dots)$. For $i = 1, \dots, N$ put

$$D_i = \partial_i + \beta \sum_{j \neq i} \frac{1 - s_{ij}}{x_i - x_j}. \quad (21.7)$$

Two remarks before anything is proved. First, D_i maps polynomials to polynomials: $(1 - s_{ij})f$ vanishes on $x_i = x_j$ and is therefore divisible by $x_i - x_j$, so no pole survives. Second, and this is where the hour is lost if it is not said, s_{ij} is an operator, not a number. It does not commute past a function, so D_i is not a derivation: for $N = 2$,

$$D_1(x_1 x_2^2) - (D_1 x_1) x_2^2 - x_1 D_1(x_2^2) = \beta (x_1 - x_2)(x_1 + x_2) \neq 0 \quad (21.8)$$

· *checked*, from $D_1 x_1 = 1 + \beta$, $D_1(x_2^2) = -\beta(x_1 + x_2)$ and $D_1(x_1 x_2^2) = x_2^2 - \beta x_1 x_2$. Everything below is written with that in mind.

Example 21.7 ($N = 2$, entirely). Write $s = s_{12}$, $x = x_1 - x_2$ and $A = \beta(1 - s)/x$, so that $D_1 = \partial_1 + A$ and $D_2 = \partial_2 - A$. Then

$$[D_1, D_2] = [\partial_1, -A] + [A, \partial_2] = -[\partial_1 + \partial_2, A].$$

Now $\partial_1 + \partial_2$ commutes with s , because exchanging the arguments does not change the total derivative, and with multiplication by $1/x$, because x is translation invariant. So $[D_1, D_2] = 0$. The whole of the commutativity theorem, at $N = 2$, is the translation invariance of $x_1 - x_2$.

Theorem 21.8 (Commutativity). $[D_i, D_j] = 0$ for all i, j , on polynomials.

Proof (★). Everything follows from one commutator with a coordinate. Writing δ for the Kronecker symbol,

$$[D_i, x_l] = \delta_{il} + \beta \sum_{p < q} (\delta_{ip} - \delta_{iq})(\delta_{lp} - \delta_{lq}) s_{pq} \quad (21.9)$$

· *checked*: for $i = l$ the sum is $\sum_{k \neq i} s_{ik}$, and for $i \neq l$ only the pair $\{i, l\}$ survives, with coefficient -1 . Next, relabelling gives $s_{pq} D_m s_{pq} = D_{s_{pq}(m)}$, whence $[s_{pq}, D_j] = -(\delta_{jp} - \delta_{jq})(D_p - D_q) s_{pq}$ and

$$[[D_i, x_l], D_j] = -\beta \sum_{p < q} (\delta_{ip} - \delta_{iq})(\delta_{lp} - \delta_{lq})(\delta_{jp} - \delta_{jq})(D_p - D_q) s_{pq} \quad (21.10)$$

· *checked*. The right-hand side is symmetric under $i \leftrightarrow j$, since i and j enter only through the factors $(\delta_{ip} - \delta_{iq})$ and $(\delta_{jp} - \delta_{jq})$. By the Jacobi identity, therefore, $[[D_i, D_j], x_l] = [[D_i, x_l], D_j] - [[D_j, x_l], D_i] = 0$ for every l . So $C = [D_i, D_j]$ commutes with multiplication by every coordinate, hence by every polynomial, and $Cf = C(f \cdot 1) = f C(1) = 0$ because $D_i 1 = 0$ · *cited: Dunkl (1989); this argument is Etingof (2009) §6.1.* ■

The first charge is already familiar: since $(1 - s_{ij})/x_{ij} + (1 - s_{ji})/x_{ji} = 0$, the deformation cancels pairwise and

$$\sum_i D_i = \sum_i \partial_i = iP, \quad P = -i \sum_i \partial_i, \quad (21.11)$$

the total momentum, undeformed · *checked*. The second charge is the Hamiltonian, and that is the content of the next theorem. Write Λ^{sym} for the symmetric polynomials.

Theorem 21.9 (The symmetric restriction). For $f \in \Lambda^{\text{sym}}$,

$$-\frac{1}{2} \sum_i D_i^2 f = \Delta^{-\beta} \left(H_C|_{v=0} \right) \Delta^\beta f = -\frac{1}{2} \sum_i \partial_i^2 f - \beta \sum_{i < j} \frac{\partial_i f - \partial_j f}{x_i - x_j}. \quad (21.12)$$

In particular $-\frac{1}{2} \sum_i D_i^2$ preserves Λ^{sym} and sends its homogeneous component of degree d into degree $d - 2$, possibly to zero.

Proof (★). Both sides in six lines. For symmetric f we have $(1 - s_{ij})f = 0$, so $D_i f = \partial_i f$; but $\partial_i f$ is not symmetric, so the second D_i must be applied honestly. Using $s_{ij}\partial_i = \partial_j s_{ij}$ and $s_{ij}f = f$,

$$D_i^2 f = D_i(\partial_i f) = \partial_i^2 f + \beta \sum_{j \neq i} \frac{\partial_i f - \partial_j f}{x_i - x_j}.$$

Summing over i doubles each unordered pair, which gives the right-hand side of (21.12).

For the middle expression, write $u = \Delta^\beta f$. Since $\partial_i \Delta^\beta / \Delta^\beta = \beta \sum_{j \neq i} (x_i - x_j)^{-1}$,

$$\Delta^{-\beta} \left(-\frac{1}{2} \sum_i \partial_i^2 \right) u = -\frac{1}{2} \sum_i \partial_i^2 f - \beta \sum_i \sum_{j \neq i} \frac{\partial_i f}{x_i - x_j} + \left(\Delta^{-\beta} \left(-\frac{1}{2} \sum_i \partial_i^2 \right) \Delta^\beta \right) f,$$

and the last bracket is $-\beta(\beta - 1) \sum_{i < j} x_{ij}^{-2}$ by the last line of Theorem 21.4, which the potential cancels. The surviving double sum is again $\beta \sum_{i < j} (\partial_i f - \partial_j f)/(x_i - x_j)$. The two expressions agree.

Finally D_i is homogeneous of degree -1 — both ∂_i and $(1 - s_{ij})/(x_i - x_j)$ lower the degree of a homogeneous polynomial by one — so $\sum D_i^2$ lowers it by two, or annihilates it; and a permutation w relabels derivatives, differences and transpositions together, so $w D_i w^{-1} = D_{w(i)}$ and $\sum_i D_i^2$ is S_N -invariant, hence preserves Λ^{sym} . *checked at $N = 3, N = 4$.* ■

Corollary 21.10 (Commuting charges, with no transfer matrix). Put $I_k = \sum_i D_i^k$ restricted to Λ^{sym} . Each I_k preserves Λ^{sym} , and $[I_k, I_m] = 0$ for all k, m . Here $I_1 = \mathbf{i}P$ is the total momentum (21.11) and $-\frac{1}{2}I_2$ is the gauge-transformed Calogero Hamiltonian (21.12).

Proof. Each I_k is S_N -invariant, so it preserves Λ^{sym} by the argument just given; and $[I_k, I_m] = 0$ on all polynomials, by Theorem 21.8. Restriction to an invariant subspace is a homomorphism, so the restrictions commute too. ■

That is a complete construction of a commuting family — infinitely many operators, N of them algebraically independent (the reading) — with no R -matrix, no transfer matrix and no ansatz. Of § 13.5’s three criteria this lecture therefore uses neither (T) nor (P) — the trapped spectrum below is as far from Poisson as a spectrum gets — but supplies the charges directly, as Lecture 14 did, and meets (S) in the asymptotic sense of § 21.6.

21.4 The trapped spectrum

Now put the trap back. The remarkable fact is that the whole spectrum follows from Theorem 21.9 alone — Theorem 21.8 is not needed for it.

One piece of notation, used from here to the end. For a partition $\kappa_1 \geq \dots \geq \kappa_N \geq 0$ the monomial symmetric polynomial m_κ is the sum of the distinct monomials $x_1^{\mu_1} \dots x_N^{\mu_N}$ with μ a rearrangement of κ : for instance $m_{(2,1,0)} = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_1 + \dots$, six terms. The m_κ with

$|\kappa| = n$ are a basis of the homogeneous symmetric polynomials of degree n , so that space has dimension the number of partitions of n into at most N parts.

Theorem 21.11 (The trapped spectrum). On Λ^{sym} ,

$$(\psi_0^C)^{-1} H_C \psi_0^C = E_0^C + \nu \sum_i x_i \partial_i - \frac{1}{2} \sum_i D_i^2. \quad (21.13)$$

Consequently the eigenvalues of H_C on $\psi_0^C \Lambda^{\text{sym}}$ are

$$E = E_0^C + \nu n, \quad n = 0, 1, 2, \dots, \quad (21.14)$$

and the multiplicity of $E_0^C + \nu n$ is the number of partitions of n into at most N parts.

Proof. Write $G = e^{-\nu \sum x_j^2/2}$ and let $h = \Delta^\beta f$ with f symmetric. From $\partial_i^2(Gh) = G(\partial_i^2 h - 2\nu x_i \partial_i h - \nu h + \nu^2 x_i^2 h)$,

$$G^{-1} \left(-\frac{1}{2} \sum_i \partial_i^2 + \frac{\nu^2}{2} \sum_i x_i^2 \right) (Gh) = -\frac{1}{2} \sum_i \partial_i^2 h + \nu \sum_i x_i \partial_i h + \frac{\nu N}{2} h,$$

the trap having cancelled. Adding the pair potential and conjugating by Δ^β , the first term becomes $-\frac{1}{2} \sum D_i^2 f$ by Theorem 21.9, while the Euler operator picks up the degree of Δ^β : $\Delta^{-\beta} \sum x_i \partial_i (\Delta^\beta f) = \sum x_i \partial_i f + \frac{1}{2} \beta N(N-1)f$. The two constants assemble into E_0^C , which is (21.13) · checked at $N = 2, 3$.

The operator on the right is triangular in the degree: $\sum x_i \partial_i$ multiplies a homogeneous polynomial of degree d by d , and $-\frac{1}{2} \sum D_i^2$ sends degree d into degree $d-2$ (Theorem 21.9), possibly to zero. So on the space of symmetric polynomials of degree at most d the matrix is block-triangular, the block of degree n carrying the scalar νn ; different blocks carry different scalars, so there is no Jordan obstruction and the lower-degree part of each eigenvector can be solved for recursively. The eigenvalues are νn , $n \leq d$, each of multiplicity the dimension of the homogeneous symmetric polynomials of degree n — the partitions of n into at most N parts. Letting $d \rightarrow \infty$ gives (21.14).

That these are *all* the eigenvalues is a density statement: the measure $|\Delta|^{2\beta} e^{-\nu|x|^2} dx$ has finite exponential moments, so polynomials are dense in its L^2 · cited. ■

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Remark 21.12 (The classical face). Corollary 12.13 says the classical trapped positions are the eigenvalues of $Q(0) \cos \nu t + \nu^{-1} L(0) \sin \nu t$, so *every* orbit returns after $2\pi/\nu$, whatever its energy and whatever the coupling. The quantum statement is (21.14): the spectrum is $E_0^C + \nu \mathbb{Z}_{\geq 0}$, so $e^{-iH_C t}$ at $t = 2\pi/\nu$ is a phase times the identity, and *every* state returns. Neither is a limit of the other; they are two theorems in the same role (Figure 21.4), which is what the Dictionary box calls a type-A row. Export E3, Lecture 5's $\oint p \, dq = 2\pi\hbar(n + \frac{1}{2})$, is exact here too, once the inverse-square coupling carries the Langer value $(\beta - \frac{1}{2})^2$ that a radial problem always needs — for every $\beta \geq \frac{1}{2}$. Comparing it with the *bare* coupling needs $\beta > 1$, where the bare orbit exists at all; the tutorial does the arithmetic.

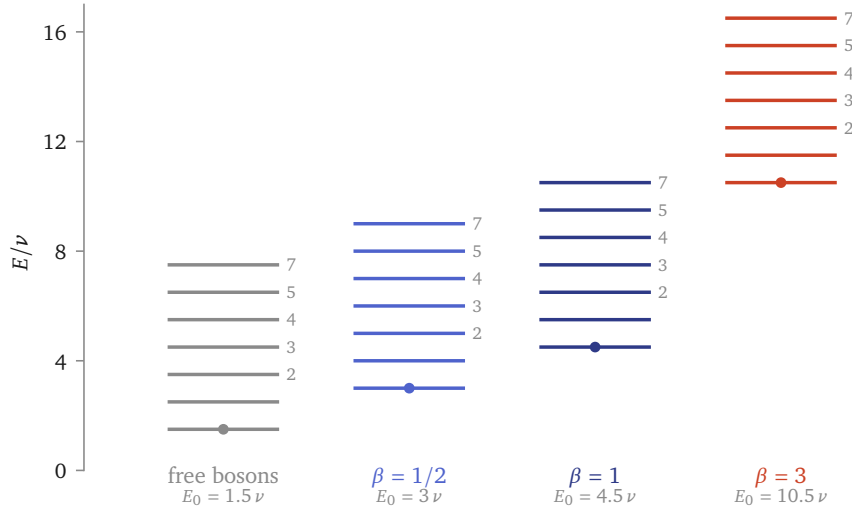


Figure 21.3: The trapped spectrum at $N = 3$ for $\beta = \frac{1}{2}, 1, 3$, beside the free-boson trap $E = \nu(N/2 + n)$. Only E_0 moves with β : the ladder above it is the same progression of spacing ν , with the same multiplicities (printed where they exceed one), the partitions $1, 1, 2, 3, 4, 5, 7$ of $n = 0, \dots, 6$ into at most three parts. The generator builds the matrix of (21.13) in exact rational arithmetic, checks that it is degree-triangular with diagonal $\nu \cdot \text{degree}$, and recovers the $\beta = 1$ ladder from a brute-force count of three free fermions in an oscillator.

21.5 Sutherland's excited states

On the circle the natural variable is not x but

$$z_j = e^{2ix_j}, \quad (21.15)$$

which has period π , as the model does. Since $z_i - z_j = 2i e^{i(x_i+x_j)} \sin(x_i - x_j)$ and $|z_j| = 1$, the ground state (21.5) is $\prod_{i<j} |z_i - z_j|^\beta$ up to the constant $2^{-\beta N(N-1)/2}$ — a constant, not a phase — and $-\frac{1}{2} \sum \partial_{x_j}^2 = 2 \sum_j (z_j \partial_{z_j})^2$ · checked. One more piece of notation: κ dominates μ (same weight) when $\kappa_1 + \dots + \kappa_r \geq \mu_1 + \dots + \mu_r$ for every r . It is a *partial* order: at $N = 3$ and weight six, $(4, 1, 1)$ and $(3, 3, 0)$ are incomparable.

Theorem 21.13 (The Sutherland spectrum). Conjugated by ψ_0^S and written in z , the operator (21.2) preserves the degree and is triangular in the dominance order on the $m_\kappa(z)$: its eigenfunctions are $\psi_0^S J_\kappa$, one for each partition $\kappa_1 \geq \dots \geq \kappa_N \geq 0$, with $J_\kappa = m_\kappa + \text{lower}$, and

$$E_\kappa = \frac{1}{2} \sum_{j=1}^N k_j^2, \quad k_j = 2\kappa_j + \beta(N+1-2j). \quad (21.16)$$

The total momentum is $\sum_j k_j = 2|\kappa|$.

Proof · sketched (★). Writing $W = \ln \psi_0^S$ in the variables z and $A_j = z_j \partial_{z_j}$, the conjugated operator is $2 \sum_j (A_j + w_j)^2$ plus the potential, where $w_j = A_j W$ acts by multiplication and is the rational function $\beta \sum_{k \neq j} z_j / (z_j - z_k) - \beta(N-1)/2$. Expanding, the terms that would produce poles cancel against the potential — exactly as in Theorem 21.4, which is the $\kappa = 0$ case — and what is left is degree-preserving. Acting on m_κ it returns $E_\kappa m_\kappa$ plus multiples of

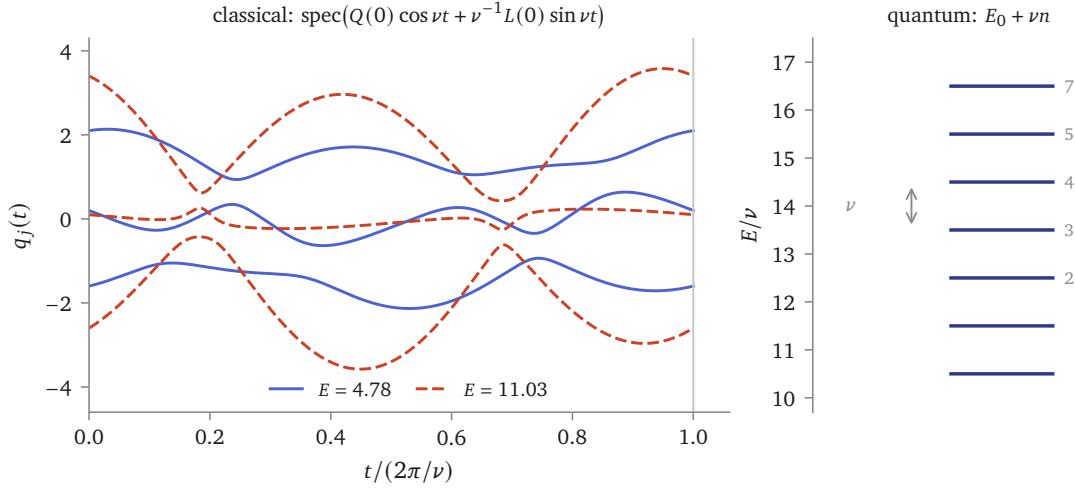


Figure 21.4: Left: two classical trajectories of the trapped rational model at $N = 3$, of energies 4.78 and 11.03 in units $\nu = g = 1$; both close at $t = 2\pi/\nu$, marked. The first comes from [Corollary 12.13](#) and agrees with a direct integration to 10^{-8} ; the second is integrated directly. Right: the quantum ladder (21.14) at $\beta = 3$, spacing ν . One period, one level spacing.

m_μ with μ strictly below κ in dominance. The coefficients of those m_μ are then solved for recursively, dividing only by the differences $E_\kappa - E_\mu$ that [Remark 21.14](#) says are non-zero. Verified symbolically for $N = 2, 3$ and $|\kappa| \leq 4$, in β , in `check-l21-algebra.py`. ■

Remark 21.14 (Why “distinct eigenvalues” is the wrong hypothesis). The recursion needs $E_\kappa \neq E_\mu$ only for μ strictly below κ , and that much is true: $E_\kappa = 2 \sum_j \kappa_j^2 + 2\beta \sum_j \kappa_j(N+1-2j) + E_0^S$ increases strictly along dominance for $\beta > 0$, since both sums do. Asking instead for *distinct* diagonal entries would be asking for something false: the incomparable $(4, 1, 1)$ and $(3, 3, 0)$ share $E_\kappa = 4\beta^2 + 12\beta + 36$ for every β · *checked*. The J_κ are the *monic* Jack polynomials — Macdonald’s $P_\kappa^{(1/\beta)}$, which he characterises by this triangularity *together with* orthogonality; his $J_\kappa^{(\alpha)}$ is a different normalisation · *cited: Macdonald §VI.10, (10.13)–(10.14); Stanley (1989), which warns about exactly this degeneracy*.

Put $\kappa = 0$ and $\sum_j (N+1-2j)^2 = N(N^2-1)/3$ returns E_0^S : the ground state is the empty partition, and the shift is the whole of its energy. The circle also admits Laurent polynomials, $\kappa_N < 0$; multiplying by $(z_1 \cdots z_N)^m$ sends $k_j \mapsto k_j + 2m$, a Galilean boost at a different energy and momentum, and $\kappa_N = 0$ picks one representative of each boost orbit.

A warning about the standard reference. *Beautiful Models* §§5.2.5–5.2.6 prints the shift twice as large as (21.16), in contradiction with its own ground-state energy; the reading says exactly where.

Free fermions with shifted momenta

The k_j of (21.16) are a set of quasi-momenta and $E = \frac{1}{2} \sum k_j^2$ is the free-particle energy. The interaction lives entirely in the shift $\beta(N+1-2j)$, which pushes neighbouring momenta at least 2β apart whatever the state ([Figure 21.5](#)).

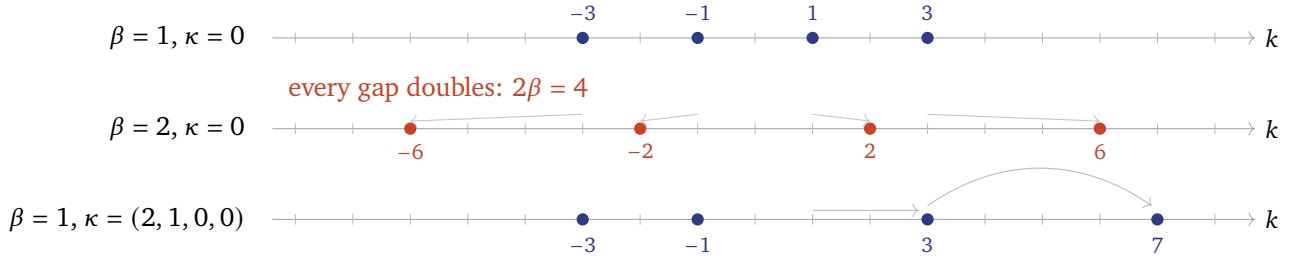


Figure 21.5: The quasi-momenta (21.16) for $N = 4$. Top: the ground state at $\beta = 1$, where the shift alone spreads the four particles over the odd integers $-3, -1, 1, 3$ — free fermions, as Example 21.3 said. Middle: raising β dilates the same pattern, gap 2β . Bottom: an excited state, $\kappa = (2, 1, 0, 0)$ at $\beta = 1$, which moves two quantum numbers up without changing the shift. Only κ is dynamical; the spreading is not.

Proposition 21.15 (An asymptotic Bethe ansatz, exactly). Put $I_j = \kappa_j + \frac{1}{2}(N+1-2j)$. Then $I_1 > I_2 > \dots > I_N$ lie in $\mathbb{Z} + \frac{N-1}{2}$ — the quantum numbers of Lecture 16, (16.11) — and $\kappa \mapsto I$ is a bijection onto the strictly decreasing sequences there that are bounded below by $-\frac{N-1}{2}$. On a circle of circumference $L = \pi$ the momenta (21.16) are the unique solution in the ordered sector $k_1 > \dots > k_N$ of

$$k_j L + \sum_{l \neq j} \varphi(k_j - k_l) = 2\pi I_j, \quad \varphi(k) = -\pi(\beta - 1) \operatorname{sgn} k, \quad (21.17)$$

that is, of Lecture 19's unified logarithmic form (19.5) with a phase function that is a step. The ground state $\kappa = 0$ is the symmetric block $I = (\frac{N-1}{2}, \dots, -\frac{N-1}{2})$, as it is for Lieb–Liniger.

Proof (★). In the ordered sector $\sum_{l \neq j} \operatorname{sgn}(k_j - k_l) = (N-j) - (j-1) = N+1-2j$ is fixed, so (21.17) reads $\pi k_j - \pi(\beta-1)(N+1-2j) = 2\pi \kappa_j + \pi(N+1-2j)$, which determines k_j uniquely and gives (21.16); and those k_j are ordered, since $k_j - k_{j+1} = 2(\kappa_j - \kappa_{j+1}) + 2\beta \geq 2\beta > 0$ — checked. The sector is a hypothesis, not a convenience: unlike Lecture 16's increasing Lieb–Liniger phase, a step does not force the order of the roots from the order of the quantum numbers, and (21.17) can have further solutions outside the sector. At $N = 2$, $\beta = 3$, $I = (\frac{1}{2}, -\frac{1}{2})$ the intended root is $(3, -3)$, of energy 9; $(-1, 1)$ solves the same two equations, with energy 1. (At $N = 2$ the reversed sector gives $k = (2-\beta, \beta-2)$, consistent only for $\beta > 2$; a spurious root needs a large enough β , not merely $\beta > 1$.) ■

The normalisation of φ is not free, and the check is Lecture 19's. The kernel is $K = \varphi' = -2\pi(\beta-1)\delta$, so the density equation (19.5) collapses to $\rho + \rho_h = \frac{1}{2\pi} - (\beta-1)\rho$; in the ground state $\rho_h = 0$ and $\rho = 1/2\pi\beta$, the direct count, the k_j being 2β apart in a box of length π . The same momenta also solve (21.17) with the step $-\pi\beta \operatorname{sgn} k$ and $I_j = \kappa_j$ — Sutherland's form, whose θ sits on the other side of the equation — but those quantum numbers are not distinct (in the ground state, all zero), so Lecture 19's counting function, which wants one root per vacancy, does not apply, and substituting its kernel anyway returns $\rho = 1/2\pi(1+\beta)$ — checked. The two differ by the statistics phase $\pi \operatorname{sgn} k$, and the course's bookkeeping — distinct quantum numbers in $\mathbb{Z} + \frac{N-1}{2}$, as in Lecture 16 — picks the first. Its vanishing at $\beta = 1$ is Example 21.3 again.

So the Bethe ansatz is present as an accounting identity and absent as a method. The scattering reading is Sutherland's, and it belongs to the model on a *line*, where the same

momentum-independent phase shift appears: a packet with non-zero relative momentum stays on one side of the step, so $d\varphi/dk = 0$ there and [Lecture 16](#)’s dictionary row gives a vanishing time delay — the quantum form of [Theorem 11.8](#)’s permuted momenta and *no displacement* · *cited: Sutherland (1971)*. On the circle there is no asymptotic region, and (21.17) is bookkeeping.

Dyson’s three ensembles

Remark 21.16 (The ground state is a random-matrix ensemble). In the variable (21.15), $|\psi_0^S|^2 \propto \prod_{i<j} |z_i - z_j|^{2\beta}$ on the unit circle, which up to normalisation is the joint eigenvalue density of Dyson’s circular ensemble of index 2β · *cited: Dyson (1962); Mehta ch. 11*. The three classical values $\beta = \frac{1}{2}, 1, 2$ are the orthogonal, unitary and symplectic ensembles, Dyson’s 1, 2, 4 — the three Wigner–Dyson curves [Figure 13.4](#) drew for *level* statistics, appearing here as the *positions* of particles in a ground state ([Figure 21.6](#)). The repulsion exponent is the Jastrow exponent: $P(s) \sim s^{2\beta}$ as $s \rightarrow 0$ is the vanishing of ψ_0^S at a collision, squared.

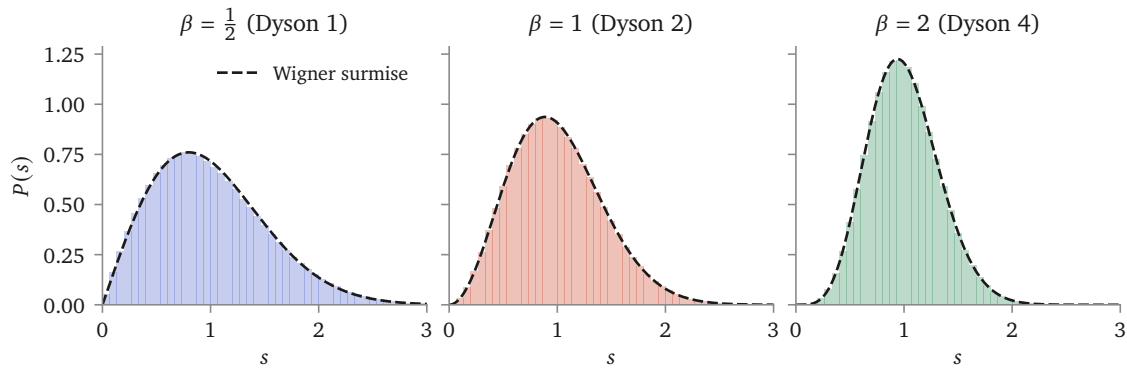


Figure 21.6: Nearest-neighbour spacings of the particle positions in the Sutherland ground state at $N = 16$, sampled directly from $|\psi_0^S|^2$ by Metropolis — no random matrix is diagonalised — against the Wigner surmises for Dyson index 2β (dashed). The generator enforces agreement to 0.008 in distance between the distribution functions on a fixed grid, a fitted small- s exponent equal to 2β within 0.3, and the same distance, at $\beta = 1$, to the spacings of an actual CUE matrix.

21.6 The dictionary: what is shared, and what is not

Nothing in this lecture was an R -matrix: no auxiliary space, no monodromy matrix, no RTT relation, and [Corollary 21.10](#) is not $\{d^k \ln t(\lambda)\}$ for any t — [Lecture 17](#)’s tower comes from one transfer matrix and one solution of the Yang–Baxter equation, while I_k comes from N operators that commute before anything is traced. Nor was there a Bethe ansatz: [Proposition 21.15](#) reports the answer in Bethe form rather than obtaining it. Both are claims about the *route*; the trigonometric model does carry a braid-group structure, Cherednik’s, and the reading says where.

What is shared: infinitely many commuting charges, one of them the Hamiltonian; a many-body problem fixed by two-body data, one exponent β per pair, with free particles at

shifted momenta; and non-diffractive scattering — criterion (S) of § 13.5 — in the strongest form there is, the phase shift being constant. The difference is where the structure sits: in [Lecture 17](#) and [Lecture 18](#) in an algebra of operators on an auxiliary space, here in the symmetric group.

Dictionary

The classical Calogero–Moser gas obtained in [Lecture 12](#) by reducing *free* motion on Hermitian matrices, the positions being the eigenvalues of a matrix moving freely $\rightarrow$ at $\beta = 1$ the Sutherland model is free fermions, and in its ground state the e^{2ix_j} are distributed exactly as the eigenvalues of a Haar-random unitary matrix ([Remark 21.16](#)) (*type A*)

Isochrony of the trapped model ([Equation \(11.10\)](#), [Corollary 12.13](#)): every orbit closes after $2\pi/\nu$, whatever its energy $\rightarrow$ the equally spaced spectrum $E_0 + \nu n$ ([Theorem 21.11](#)): every state returns after $2\pi/\nu$, whatever its energy (*type A*)

Reading

21.7 Creation operators, Cherednik’s operators, and exclusion statistics

Four continuations, each of which is a small subject.

What the charges are, and what they are not. I_k of [Corollary 21.10](#) is a differential–difference operator of order k involving every particle; it is not a sum of local densities of bounded range, and there is no lattice here for [Definition 13.2](#) to apply to. The family is also not the one [Lecture 11](#) produced classically: $\text{tr } L^k$ quantises to something with ordering ambiguities, and the Dunkl construction sidesteps that question rather than answering it. What is true is that the I_k commute, that I_2 is the Hamiltonian, and that N of them are algebraically independent — which is the count Liouville would have asked for · *cited: Polychronakos (1992)*.

Creation operators. [Theorem 21.11](#) gives the trapped spectrum but not a ladder: it produces eigenvalues by triangularity, not by raising a state. A ladder exists, and it lives in the *gauge-transformed* picture, which is the thing to get right. Put

$$a_i^\pm = \frac{1}{\sqrt{2\nu}}(\nu x_i \mp D_i), \quad [a_i^-, a_j^+] = [D_i, x_j]$$

with (21.9) on the right: a deformed Heisenberg algebra whose deformation is made of transpositions. The symmetric combinations $B_k^+ = \sum_i (a_i^+)^k$ raise the energy by $k\nu$, and the N of them with $k = 1, \dots, N$ generate the whole spectrum *freely* — but from the Gaussian $G = e^{-\nu \sum_j x_j^2/2}$, not from ψ_0^C . The physical states are $\Delta^\beta B_{k_1}^+ \cdots B_{k_r}^+ G$; equivalently, the physical ladders are $\mathcal{B}_k^+ = \Delta^\beta B_k^+ \Delta^{-\beta}$, applied to ψ_0^C · *cited: Brink, Hansson and Vasiliev (1992), eqs. (2), (4), (12); Polychronakos (1992)*. Applying the displayed B_k^+ to ψ_0^C itself leaves the Hilbert space: at $N = 2$, $\beta = \nu = 1$, $B_2^+ \psi_0^C$ acquires a $2/r^2$ and is not square-integrable. With the gauge right, the spectrum is that of N independent bosonic modes of frequencies $\nu, 2\nu, \dots, N\nu$, so the multiplicity of $E_0 + \nu n$ is the number of partitions of n into parts of size at most N — the conjugate of [Theorem 21.11](#)’s count, and equal to it. That is also why the multiplicities do not depend on β .

Cherednik’s operators. The D_i of (21.7) are the *rational* Dunkl operators, and [Theorem 21.13](#) did not use them: the Sutherland model needed its own conjugation, done by

hand. There is a trigonometric analogue, Cherednik’s: commuting first-order differential–difference operators in z , built from $z_j \partial_{z_j}$ and the transpositions, whose algebra is the degenerate affine Hecke algebra — and that is where the braid relations hidden in this model live. Two distinctions are easy to lose. The individual Cherednik operators do *not* preserve the symmetric polynomials; their eigenfunctions are the *non-symmetric* Jack polynomials, and it is their symmetric combinations that act on Λ^{sym} and are diagonalised by the J_κ of [Theorem 21.13](#). And the q -difference theory, where $z_j \mapsto qz_j$, is a different and larger story: it produces Macdonald polynomials and the double affine Hecke algebra, with the Jack polynomials recovered in a degeneration · *cited: Cherednik (1991); Macdonald §VI.10*.

A discrepancy in the standard reference. Sutherland’s *Beautiful Models* §5.2.5 gives the trigonometric spectrum twice. Its first form, $E = (2\pi^2/L^2) [\sum_j n_j^2 + \lambda \sum_{j>i} |n_j - n_i|] + E_0$ on a circle of circumference L , is (21.16) exactly, with $\lambda = \beta$, $n = \kappa$, $L = \pi$. The last line of the same equation, and the momenta of §5.2.6, print $k_j = (2\pi/L)[n_j + \lambda(N+1-2j)]$ — *twice* the shift, and inconsistent with the same book’s own $E_0 = \pi^2 \lambda^2 N(N^2-1)/6L^2$. The free-fermion case settles it: at $\lambda = 1$, $N = 2$, $L = \pi$ the ground state is $\sin(x_1 - x_2)$, whose momenta are ± 1 and whose energy is 1, which is that E_0 ; the printed momenta give ± 2 and 4. Replacing λ by $\lambda/2$ in those two places repairs both · *checked; pages rendered, not extracted*.

Fractional exclusion statistics. The density equation after [Proposition 21.15](#) reads $\rho_t = \frac{1}{2\pi} - (\beta - 1)\rho$: each root present removes $\beta - 1$ vacancies beyond the one it occupies. Haldane read this as a statistics interpolating between bosons and fermions, with $\beta = 1$ the fermionic point where nothing is removed. The thermodynamics is then a one-line modification of [Lecture 20](#)’s counting — the binomial coefficient $\binom{L\rho_t d\lambda}{L\rho d\lambda}$ of [Proposition 20.1](#) acquires a β -dependent number of slots — and reproduces the Calogero–Sutherland free energy · *cited: Haldane (1991); Wu (1994)*. It is the course’s cleanest example of a *delta-function* kernel: [Lecture 19](#)’s integral equation becomes local in momentum, and the thermodynamics is algebra rather than an integral equation.

21.8 Two particles, done by hand

Everything in this lecture is visible at $N = 2$, where no symmetric-function theory is needed. Separate the centre of mass $R = (x_1 + x_2)/2$ and the relative coordinate $r = x_2 - x_1 > 0$ on the chamber. Then $-\frac{1}{2}(\partial_1^2 + \partial_2^2) = -\frac{1}{4}\partial_R^2 - \partial_r^2$ and $\frac{1}{2}\nu^2(x_1^2 + x_2^2) = \nu^2 R^2 + \frac{1}{4}\nu^2 r^2$, so (21.1) splits into an oscillator in R , of levels $\nu(n_{\text{cm}} + \frac{1}{2})$, and

$$h_{\text{rel}} = -\partial_r^2 + \frac{\beta(\beta-1)}{r^2} + \frac{\nu^2}{4}r^2,$$

whose eigenfunctions are $r^\beta e^{-\nu r^2/4}$ times Laguerre polynomials in $\nu r^2/2$ and whose levels are $\nu(2n + \beta + \frac{1}{2})$. Five things to check, in this order.

(a) *The ground state and the spectrum.* The product of the two ground states is $r^\beta e^{-\nu(2R^2+r^2/2)/2}$, which is (21.5) at $N = 2$, with $E_0 = \nu(\frac{1}{2} + \beta + \frac{1}{2}) = \nu(1 + \beta)$, matching (21.6). The full spectrum is $E_0 + \nu(n_{\text{cm}} + 2n)$, and the number of ways to write k as $n_{\text{cm}} + 2n$ is $\lfloor k/2 \rfloor + 1$, the partitions of k into at most two parts. [Theorem 21.11](#) at $N = 2$, obtained without a single Dunkl operator — and, unlike the theorem, with completeness thrown in, since the radial problem’s eigenfunctions are a known complete set. A finite-difference diagonalisation of h_{rel} confirms the levels and finds nothing between them.

(b) *The domain is doing work.* The other indicial root gives $r^{1-\beta}e^{-\nu r^2/4}$, also positive, also a solution, with the lower value $\nu(\frac{3}{2} - \beta)$. For $\beta \geq \frac{3}{2}$ it is not square-integrable — at $\beta = \frac{3}{2}$ exactly, r^{-1} diverges logarithmically — and the question does not arise; for $\frac{1}{2} < \beta < \frac{3}{2}$ it is, and it belongs to a different self-adjoint operator — one of the family [Definition 21.2](#) chooses among. At the endpoint $\beta = \frac{1}{2}$ the two exponents coincide, the second solution is $r^{1/2} \ln r$, and the two values are equal: there is nothing to choose between. This is [Remark 21.5](#) in one dimension, and it is the reason [Definition 21.2](#) comes before [Theorem 21.4](#). For $N \geq 3$ the partner fails earlier than $\beta = \frac{3}{2}$, at the total collision; [Remark 21.5](#) says where.

(c) *The Dunkl algebra.* With $s = s_{12}$, $D_1 = \partial_1 - \beta(1-s)/r$ and $D_2 = \partial_2 + \beta(1-s)/r$. Verify on $f = x_1^2$ that $D_1 f$ and $D_2 f$ are polynomials, that $D_1 D_2 f = D_2 D_1 f$, and reproduce (21.8) by hand — the trap of [Definition 21.6](#), worth a paragraph of warning. Then take f symmetric, check $D_i f = \partial_i f$, and rediscover (21.12): $-\frac{1}{2} \sum D_i^2 f = -\frac{1}{2} \sum \partial_i^2 f + \beta(\partial_1 f - \partial_2 f)/r$. At $N = 3$ the same calculation on $f = x_1^2$ gives $[D_1, D_2] f = 0$ in four lines, and is the smallest case in which the three-index terms of [Theorem 21.8](#) appear at all.

(d) *Bohr–Sommerfeld, exactly.* For $A > 0$ and $E > \nu\sqrt{A}$ the classical orbit of h_{rel} with coupling A has two turning points and action $\oint p \, dr/2\pi = (E/\nu - \sqrt{A})/2$, so $I = n + \frac{1}{2}$ gives $E = \nu(2n + 1 + \sqrt{A})$. That is the exact level $\nu(2n + \beta + \frac{1}{2})$ if and only if $\sqrt{A} = \beta - \frac{1}{2}$: the Langer value $A = (\beta - \frac{1}{2})^2$, not the bare $A = \beta(\beta - 1)$. Comparing the two needs $\beta > 1$, where the bare coupling is positive; for $\frac{1}{2} \leq \beta < 1$ it is negative, the classical particle falls into the collision and the bare action diverges, while the Langer-corrected rule still returns the exact levels. Where the comparison is legitimate the bare rule is wrong by $\nu(\beta - \frac{1}{2} - \sqrt{\beta(\beta - 1)})$, a constant that does not shrink at large quantum numbers — checked. Export E3 is therefore exact here in the sense semiclassics always means it, and this is the cheapest place in the course to see what the Langer correction is for.

(e) *Sutherland at $N = 2$.* On the circle, $\psi_0^S = \sin^\beta r$ and $E_0^S = \beta^2 \cdot 2 \cdot 3/6 = \beta^2$; and (21.16) gives $k_{1,2} = 2\kappa_{1,2} \pm \beta$, so $E_\kappa = \frac{1}{2}[(2\kappa_1 + \beta)^2 + (2\kappa_2 - \beta)^2]$. At $\beta = 1$, $\kappa = 0$, that is $E = 1$ with momenta ± 1 : the two impenetrable bosons of [Example 21.3](#), which are one antiperiodic fermion pair. The formal value $\beta = 0$ gives free bosons at rest, $E = 0$ — the one exponent at which the shift disappears, and the only one at which the ground state is not an ensemble. It lies outside [Definition 21.2](#)'s $\beta \geq \frac{1}{2}$, being the partner of $\beta = 1$.

Summary.

- *The models* (21.1), (21.2), in units $\hbar = m = 1$ ([Remark 21.1](#)) with coupling $\beta(\beta - 1)$; the operator is the Friedrichs extension and its eigenfunctions vanish as $|x_i - x_j|^\beta$ ([Definition 21.2](#)) — the larger indicial exponent, and a genuine part of the model.
- *Jastrow ground states* ([Theorem 21.4](#)): $E_0 = \nu(N/2 + \beta N(N - 1)/2)$ trapped, $E_0 = \beta^2 N(N^2 - 1)/6$ on the circle, both by one page of differentiation. Positivity is not what makes them ground states ([Remark 21.5](#)): for $\beta > \frac{1}{2}$ the partner function $\prod \sin^{1-\beta}$ is positive too and solves the same equation with a lower value, and the form domain is what excludes it — for $N \geq 3$ by the total collision, $\beta < 1 + 1/N$, not by any pair. At $\beta = \frac{1}{2}$ the two coincide.
- *Dunkl operators* (21.7) commute ([Theorem 21.8](#); the $N = 2$ case is the translation invariance of $x_1 - x_2$, and the general case follows in five lines from the commutator of D_i with a coordinate, in the reading), and on symmetric functions $-\frac{1}{2} \sum D_i^2$ is the gauge-transformed Calogero operator ([Theorem 21.9](#)). Hence commuting charges $I_k = \sum D_i^k$ with no transfer matrix and no ansatz ([Corollary 21.10](#)).

- The trapped spectrum (21.14) is $E_0 + \nu n$ with multiplicity $p(n \mid \text{at most } N \text{ parts})$, by triangularity in the degree (Theorem 21.11) — and Dunkl commutativity is not needed for it. Its classical face is Lecture 11’s isochrony (Remark 21.12); Bohr–Sommerfeld reproduces it exactly once the coupling carries its Langer value $(\beta - \frac{1}{2})^2$, and comparing that with the bare $\beta(\beta - 1)$ needs $\beta > 1$.
- Sutherland’s spectrum (21.16): $E = \frac{1}{2} \sum k_j^2$ with $k_j = 2\kappa_j + \beta(N + 1 - 2j)$ on the circle of circumference π — free fermions with momenta spread by 2β per pair, eigenfunctions the Jack polynomials (Theorem 21.13), whose triangularity argument needs dominance monotonicity, not distinctness. Those k_j solve Lecture 19’s unified Bethe equation (21.17) in the ordered sector, with the step $\varphi = -\pi(\beta - 1) \operatorname{sgn} k$ and Lecture 16’s quantum numbers (Proposition 21.15) — and the $\beta - 1$, not β , is fixed by the density equation, not by the spectrum. The ansatz is a result here, not a method. The standard reference prints the shift twice as large; the reading records where.
- Dyson (Remark 21.16): $|\psi_0^S|^2$ is the circular ensemble of index 2β , so $\beta = \frac{1}{2}, 1, 2$ are Dyson’s 1, 2, 4 — Lecture 13’s three curves, as a ground state.

Tutorial. Exercise Sheet 21.

References

B. Sutherland, *Beautiful Models: 70 Years of Exactly Solved Quantum Many-Body Problems*, World Scientific (2004), §§5.1–5.2 — this lecture’s §§21.2 and 21.5, and the source of the asymptotic-Bethe-ansatz reading (but see the warning in this lecture’s reading section); F. Calogero, J. Math. Phys. **12** (1971) 419 (the trapped rational model and its spectrum); B. Sutherland, Phys. Rev. A **4** (1971) 2019 and **5** (1972) 1372 (the original papers on the model on a circle); M. Girardeau, J. Math. Phys. **1** (1960) 516 (the Bose–Fermi map of Example 21.3); C. F. Dunkl, Trans. Amer. Math. Soc. **311** (1989) 167 (the operators and their commutativity), with P. Etingof, *Lectures on Calogero–Moser Systems*, arXiv:math/0606233 (2009), §6.1, whose short proof is the one printed here; A. P. Polychronakos, Phys. Rev. Lett. **69** (1992) 703, and L. Brink, T. H. Hansson and M. A. Vasiliev, Phys. Lett. B **286** (1992) 109 (the exchange-operator formalism and the creation operators of the reading); I. Cherednik, Invent. Math. **106** (1991) 411 (the trigonometric operators and the degenerate affine Hecke algebra); I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford (1995), §VI.10 (Jack polynomials and the Sutherland operator), with R. P. Stanley, Adv. Math. **77** (1989) 76; F. J. Dyson, J. Math. Phys. **3** (1962) 140, and M. L. Mehta, *Random Matrices*, 3rd ed., Academic Press (2004), ch. 11 (the circular ensembles); F. D. M. Haldane, Phys. Rev. Lett. **67** (1991) 937, and Y.-S. Wu, Phys. Rev. Lett. **73** (1994) 922 (exclusion statistics); M. Reed and B. Simon, *Methods of Modern Mathematical Physics II*, Academic Press (1975), §X.1 and its appendix (the inverse-square potential and its self-adjoint extensions). A. P. Polychronakos, J. Phys. A **39** (2006) 12793 is a review of the whole subject.